

Exact monomial moments with a constant phase (Taylor-tree Stage 2)

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Abstract

Units 227–229 complete Stage 2 of the Taylor-tree programme. The Taylor-tree terms are monomial integrals with a constant phase $a = \xi(0)$ and a phase power p , $T(N) = \int_{(0,1]^d} u^h (\sqrt{N} u^k)^p e^{-\beta N u^{2k} + \beta \sqrt{N} u^k a} du$. In the state-density variable $\tau = u^{2k}$ the kernel is $g(N\tau)$ with the *phase kernel* $g(t) = (\sqrt{t})^p e^{-\beta t + \beta \sqrt{t} a}$, so $T(N) = \prod \frac{1}{2k_i} \int_0^1 v(\tau) g(N\tau) d\tau$ with the exact density v of Stage 1. Substituting $t = N\tau$ termwise and expanding $(\log N - \log t)^j$ binomially gives the **exact identity** (Headline XXIII, `monomialPhase_eq`), for every $N > 0$,

$$T(N) = \prod_i \frac{1}{2k_i} \sum_{(\mu, j, c) \in v} c N^{-\mu} \sum_{i \leq j} \binom{j}{i} (\log N)^{j-i} \int_0^N t^{\mu-1} (-\log t)^i g(t) dt,$$

a finite identity with no Mellin inversion, no asymptotic expansion of the density and no interchange of infinite sums: the paper's substitution of the state-density expansion into the standard integral. The truncated moments differ from the full *fluctuation moments* $\int_0^\infty t^{\mu-1} (-\log t)^i g(t) dt$ by exponentially small tails: $g(t) \leq (\sqrt{t})^p e^{\beta a^2/2} e^{-\beta t/2}$ (from $2\sqrt{t}a \leq t + a^2$) and $t^m e^{-\beta t/4} \leq m!(4/\beta)^m$ give $|\int_N^\infty \dots| \leq C e^{-\beta N/4}$ (`tail_le`), whence **Headline XXIII'** (`monomialPhase_isBig0`): $T(N)$ minus the main sum with full moments is $O(e^{-\beta N/8})$ — the paper's $\Delta = O(e^{-\varepsilon n})$ for one monomial term, with the main sum a finite combination of the scales $N^{-\mu} (\log N)^k$, $\mu \in \{(h_i + 1)/(2k_i)\}$, $k < \text{multiplicity}$. Numerics ($d = 2$, $h = (0, 1)$, $k = (1, 1)$, $p = 1$, $a = 0.3$, $\beta = 1$, $N = 5$): both sides of XXIII equal 0.1692991699. Zero `sorry/axiom`; 9106 jobs.

1 The things formalised

```
def phaseKernel a p t := sqrt t ^ p * exp (-( t) + sqrt t a)
def truncMoment a p i N := t in Ioc 0 N, t^(-1) (-log t)^i g t
theorem integrableOn_truncMoment (h : 0 < ) (N) -- split (0,1] (1,N]
theorem basis_scaling (h : 0 < ) (hN : 0 < N) :
  in Ioc 0 1, powLogBasis j * g (N)
  = N^(-) * i range (j+1), C(j,i) (log N)^(j-i) * truncMoment a p i N
theorem integral_unitBox_monomial_eq_stateDensity (f 0 measurable) :
  unitBox (n+1), ( u^h) f ( u^(2k)) = ( 1/(2k)) * Ioc 0 1, eval rep z * f z
theorem phaseKernel_box_eq : (sqrt N u^k)^p e^{...} = g (N u^(2k))
theorem monomialPhase_eq (hk) (hN : 0 < N) : T N = ( 1/(2k)) * (rep.map ...).sum -- XXIII
def fluctMoment a p i := t in Ioi 0, t^(-1) (-log t)^i g t
theorem phaseKernel_le (h) : g t sqrt t^p * (exp ( a^2/2) * exp (-( t/2)))
theorem tail_integrand_le (t 1) : |...| tailConst a p i * exp (-( t/4))
theorem integrableOn_fluct (h) ; truncMoment_eq_fluct_sub (N 1)
```

```

theorem tail_le (N 1) : | Ioi N, ...| tailConst * (4/) * exp (-( N/4))
theorem log_pow_mul_exp_le : N^(-) (log N)^k e^{- N/4} k! (8/)^k e^{- N/8}
theorem term_tail_isBig0; def monomialMainSum
theorem monomialPhase_isBig0 (h) : (fun N => T N - monomialMainSum N) =0[atTop] exp (-
( N/8))

```

2 Mathematics

Everything is elementary once the density is exact. The only analytic inputs are the three bounds in the abstract and the substitution $t = N\tau$ on an interval integral. The tail constant is explicit, $C = e^{\beta a^2/2} m! (4/\beta)^m$ with $m = \lceil \nu \rceil + i + p$, and the final $O(e^{-\beta N/8})$ absorbs the factors $N^{-\mu} (\log N)^k$ of the main sum through $(\log N)^k \leq N^k \leq k! (8/\beta)^k e^{\beta N/8}$. Astra #26's warning is respected: the identity is for a *fixed* monomial and phase order; nothing is summed over infinitely many monomials here.

3 Lean notes

- `Integrable.bdd_mul (c := C) hf.aestronglyMeasurable (ae bound)` for “integrable times bounded”; ascribe the `IntegrableOn` type before `.congr_fun`.
- `Asymptotics.IsBig0.sum` takes the family as `A` and yields a Pi-sum; convert with `.congr_left fun N => Finset.sum_apply _ _ _`.
- `gcongr` on products with several factors generates nonnegativity side goals in an unexpected order; explicit `mul_le_mul_of_nonneg_left` chains are more predictable.
- Naming a hypothesis `h` when `h` is the exponent function silently shadows it.

4 Status

Stages 1–2 complete (Headlines XXII–XXIII'). Next: Stage 3 (spectral truncation, per Astra #26): finitely many exponents below a cutoff on the rational lattice, uniform bounds on the density coefficients, all phase orders summed at each retained exponent with the Gaussian domination $\sum_p (\beta B)^p / p! \int t^{\mu+p/2-1} |\log t|^j e^{-\beta t + \beta |a| \sqrt{t}} = \int t^{\mu-1} |\log t|^j e^{-\beta t + \beta (|a|+B) \sqrt{t}}$.